MIL-C-46793B(ME)
28 October 1983
SUPERSEDING
MIL-C-46793A(ME)
18 February 1976

MILITARY SPECIFICATION

CUPOLA BODY ASSEMBLY FOR ARMORED VEHICLE

BRIDGE LAUNCHER, M60A1 AND M48A5 TANK CHASSIS

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the cupola body assembly required for the armored vehicle bridge launcher, M60A1 and M48A5 tank chassis.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-529 - Enamel, Alkyd, Semi-Gloss.

MILITARY

MIL-T-704 - Treatment and Painting of Materiel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packing of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13220E8285	- Cupola Body Assembly.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125).

AMERICAN SOCIETY OF MECHNICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code for Unfired Pressure Vessels, Section IX, Welding Qualifications.

ASA B46.1	- Surface Texture.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The cupola body assembly shall be in accordance with TA13220E8285 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the cupola. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more cupola body assemblies as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified on the applicable drawings.

3.4 Vision blocks. Vision blocks shall be installed in cupolas as specified on drawings. Glass of vison blocks shall be free of sealing compounds and paint after assembly.

3.5 Surfaces. Machine surfaces shall conform to ASA B46.1.

3.6 Identification marking. All components shall be marked for identification in accordance with MIL-STD-130.

3.7 Treatment and painting. All exterior surfaces of the cupola, except the vision block surfaces, shall be treated and painted in accordance with MIL-T-704, Type A. All interior surfaces of the cupola, except the vision block surfaces, shall be treated and painted in accordance with MIL-T-704, except that the paint used shall conform to TT-E-529 matching color No. 24491 of FED-STD-595.

3.8 Workmanship.

3.8.1 Castings. All parts, components, and assemblies of the cupola which includes castings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechancial properties or affect function.

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3.8.2 Metal fabrication. Metal used in fabrication shall be free from kinks or sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. All bends shall be made with controlled means to insure uniformity of size and shape. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.8.3 Welding and welders.

3.8.3.1 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisons of the American Welding Society publication "D1.1, Structural Welding Code". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.8.3.2 Welders. Before assigning any welder to manual welding work covered by this specification the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The cupola body assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each cupola body assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The cupola body assembly shall be examined for the following defects:

101. Nonconformance to the dimensions specified on the drawings.
102. Vision of personnel inside of cupola impaired by defective vision blocks or by foreign matter on the vision block surfaces.
103. Components missing or not as specified.
104. Identification marking missing or illegible.
105. Treatment and painting not as specfied. Presence of blisters, thin spots, lumps, scratches, or runs.
106. Loose, damaged, defective, binding, improperly installed, or interfering fits of screws, wires, vision blocks, and latch components.

4.6 Inspection of packaging. The preservation, packing and marking shall be examined to determine compliance with the requirements of MIL-STD-1188.

5. PACKAGING

5.1 Preservation, packing and marking (see 6.4). Each complete cupola body assembly shall be preserved, packed, and marked in accordance with MIL-STD-1188.

5.2 Packing (see 6.4). Each complete cupola body assembly shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings.

5.3 Marking (see 6.4). Marking shall be in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The cupola body assemblies are intended for use as components of vision cupolas for the driver and commander of the armored vehicle bridge launcher, M60A1 and M48A5 tank chassis.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDiss applicable and exceptions thereto (see 2.1.1).
c. Time frame required for submission of the preproduction model, and number required (see 3.2).

6.3 Preproduction model. Any changes or deviations of cupolas from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish cupola body assemblies to this specification.

6.4 Packaging. Levels of packaging, and packing have not been included as it is not anticipated that the cupola body assemblies will require shipment other than from the contractor to the Armored Vehicle Bridge Launcher manufacturer and will not encounter the hazards defined for overseas shipment. For purposes of preservation/packing level markings, the marking shall be C/C.

Custodian:
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-46793B(ME)	Cupola Body Assembly for Armored Vehicle Bridge Launcher, M60A1 and M48A5 Tank Chassis

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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